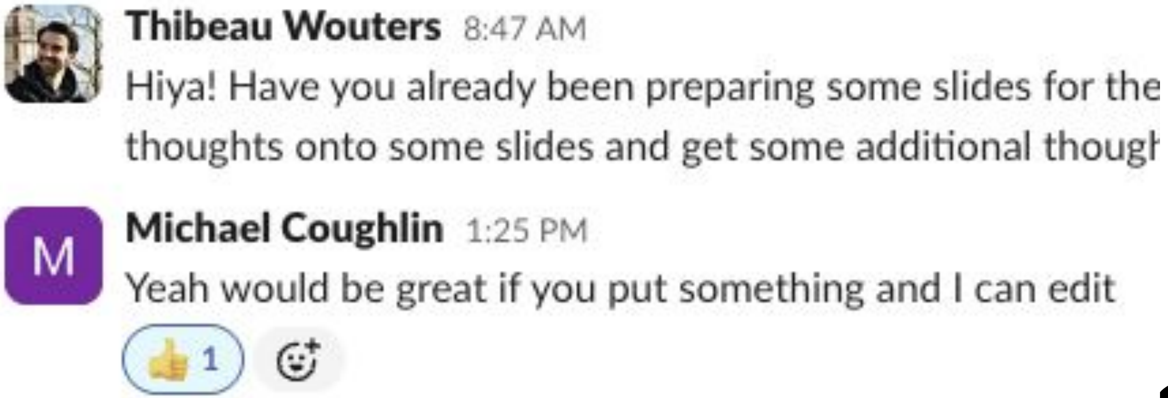
“Claude can do that!”

Or: *AI in NMMA*

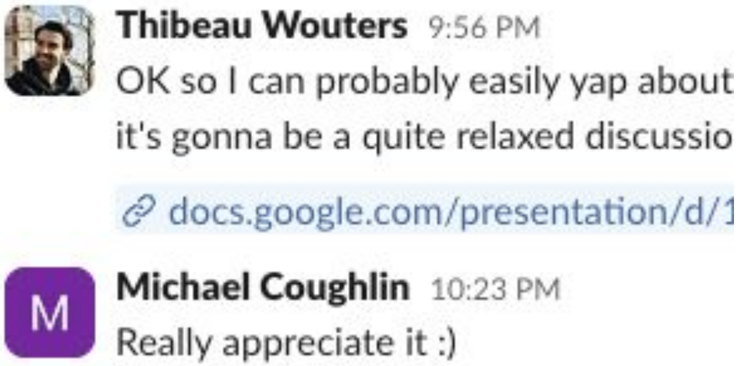
By: Thibeau Wouters, Michael Coughlin



Claude can do that!”

Or: *AI in NMMA*

By: Thibeau Wouters, Michael



Outline

- “AI”?
- Thoughts/examples on using AI for
  - Developing code and working with code
  - Research
  - Writing up your research
- Summary/Closing thoughts
- What can it do for NMMA retreat (and beyond)

AI

“AI”?

- Here does **not** refer to
  - Surrogate KN models
  - Normalizing flows
  - Foundational models
  - Simulation-based inference

- Expect those in respective fiesta, statistics,... sessions

- Here: AI for code development

Henrik Rose <henrik.rose@uni-potsdam.de> Wed 22 Jul, 15:35

to natalie.williams, Michael, Mattia, Hauke, Peter, anna.puecher, T.R.I., Tim

Dear all,

You should just have received an email because I have added you to a github "retreat" repo.

As Fahren is less than two months away, I write to you now (if I have not done so already for some of you) because I would like you to contribute a roughly 60 (up to 90) minutes "hands-on" tutorial session in Fahren that we hope will improve either the code or the understanding of its outputs.

Below you find what tutorial I envision you to provide. **Please confirm by 31 July (again)** or, if you do not see yourself equipped to do this, please convince another participant to take over the task:)

1. Michael: How to write a good test with pytest? How to write a good github workflow? (+Thibeau? It might be fine to split this into two shorter sessions)

Henrik Rose <henrik.rose@uni-potsdam.de>

to T.R.I., Michael

Hi,

both of you have responded yesterday with essentially the same statement: Claude could do that.

AI

- Agentic coding: large language model (LLM) agents plan, develop, execute and test code (or more generally, text)
  - Claude Code (Anthropic)
  - Codex (OpenAI)
  - Gemini CLI (Google, Alphabet)

- LLM chat with access to local project files, Python environment, SSH to cluster,..

- Not free (Pro membership: 20 euros/month)

- Limited usage per 5 hours, and per week – but plenty at Pro as individual

Disclaimer

- AI is a disrupting tool on individual and global level, use with care
- I fully understand anyone being against AI tool, for the individual/global reasons
- I believe they can be useful tools in accelerating parts of our workflow and making science more efficient.
- But any tool is dangerous if you don’t know how to use it properly
- I am not sure if I know how to use it properly, but I learn every day, I’ll do my best to share best practices/lessons learnt (in this presentation and always)

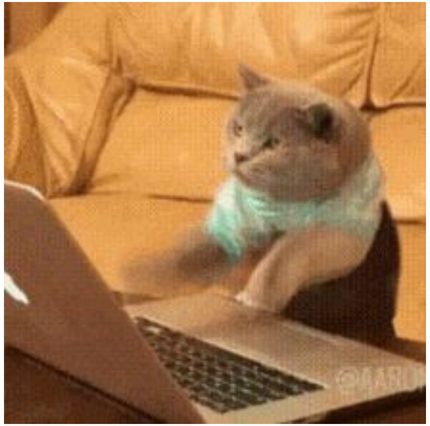
Claude Code vs Claude Chat

- Chats: quick single less technical questions, advanced web search,...
- Claude Code: can
  - Read locally cloned Git repos: full source code
  - Read locally downloaded files (eg arXiv paper source code)
  - Write to files (Python scripts, TeX summary)
  - SSH into clusters to check data, submit jobs,...

Why should you consider AI for coding/research?

- AI agents process text (input/output) **much faster**
- Open source code = thousands of lines of text. Are you gonna read each and every line?

With AI



Without AI

Why should you consider AI for coding/research?

- AI is becoming extremely **smart extremely fast***
- OpenAI disproof of unit distance problem, with thoughts from mathematicians

The unit distance problem was the only problem from discrete geometry I included; the distinct distance problem was also a strong contender, but since this has been (almost) completely solved by Guth and Katz, the unit distance problem was a better example of a still open yet natural question that has resisted proof for decades.

While I believed that AI would make some progress on at least a couple of the problems in that list eventually, I did not expect this to happen just one month later!

*: see few slides further down

Why should you consider AI for coding/research?

- AI makes research **parallel**
- Dig into research rabbit holes to test the waters

On examining the construction, it becomes more clear how people had missed this before – it requires the confluence of several different unlikely events: that a good mathematician is

(1) spending significant time in thinking about the unit distance conjecture in the first place;
(2) seriously trying to disprove it, despite the oft-repeated belief of Erdős that it is true;
(3) believes that there is mileage in generalising the original construction to other number fields, and so is willing to expend significant time in exploring such constructions[3]; and
(4) sufficiently familiar with the relevant parts of class field theory to recognise that the appropriately phrased question about infinite towers of number fields with appropriate parameters can be solved using existing theory.

The AI met all of these criteria, and its success here echoes previous achievements: it often produces the most surprising results by persevering down paths that a human may have dismissed as not worth their time to explore, combining superhuman levels of patience with familiarity with a vast array of technical machinery.

Why should you consider AI for coding/research?

- AI makes research **parallel**
- Dig into research rabbit holes to test the waters

On examining the construction, it becomes more clear how people had missed this before – it requires the confluence of several different unlikely events: that a good mathematician is

(1) spending significant time in thinking about the unit distance conjecture in the first place;
(2) seriously trying to disprove it, despite the oft-repeated belief of Erdős that it is true;
(3) believes that there is mileage in generalising the original construction to other number fields, and so is willing to expend significant time in exploring such constructions[3]; and
(4) sufficiently familiar with the relevant parts of class field theory to recognise that the appropriately phrased question about infinite towers of number fields with appropriate parameters can be solved using existing theory.

The AI met all of these criteria, and its success here echoes previous achievements: it often produces the most surprising results by persevering down paths that a human may have dismissed as not worth their time to explore, combining superhuman levels of patience with familiarity with a vast array of technical machinery.

September 8, 2026 Research Publication

On the Navier–Stokes Millennium Prize Problem

Read the paper

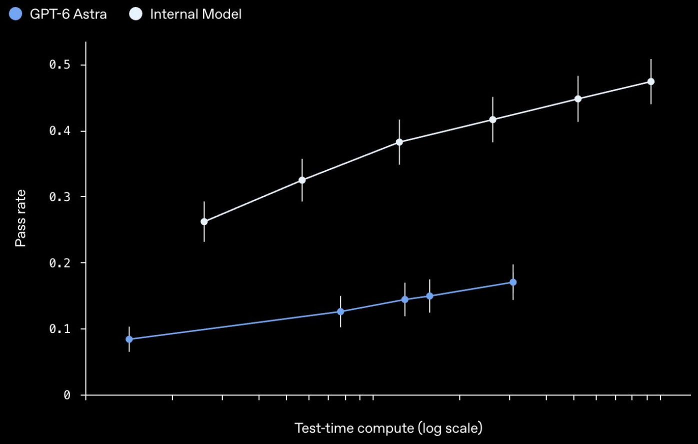
Link to Lean formalized proof >

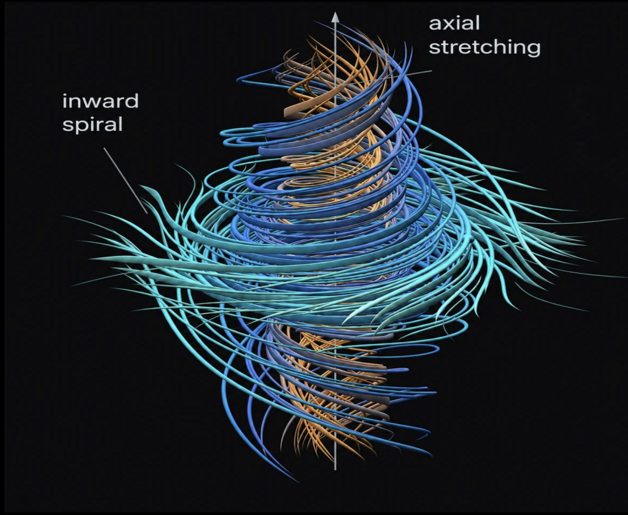
Listen to article 8:43

Share

We're sharing a solution to the Navier–Stokes existence and smoothness problem, one of the Millennium Prize Problems. This proof, produced by an internal OpenAI system, shows that the dynamics of the Navier-Stokes equations for fluid motion can develop a singularity in finite time. We're sharing both a writeup of the proof and a formalization in Lean.

Performance of GPT-6 Astra and our Internal Model on a curated set of open math problems





OpenAI

- But…
  - Internal model: closed science?
  - What did we learn?
  - Who's gonna verify it?

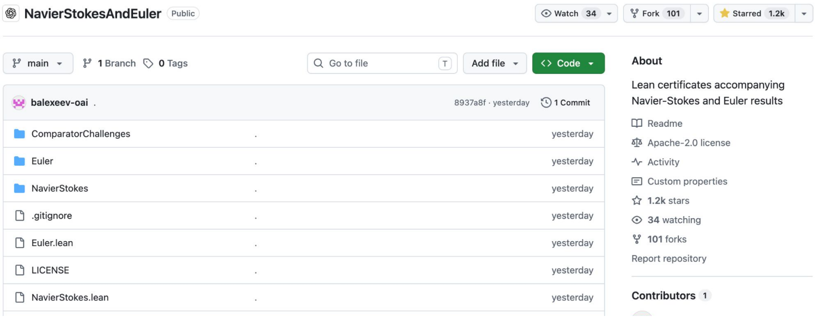
FINITE TIME BLOWUP FOR NAVIER–STOKES

OPENAI

Abstract. For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier–Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy.

Contents

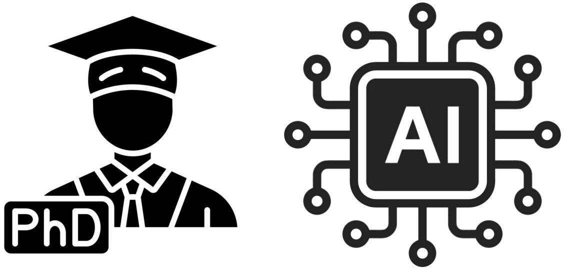
For each problem, we prompted different groups of agents with different variants of the problem statement, covering all variants of the problem. For the Navier–Stokes problem, we suggested versions "A" and "B" (particular forms of the Navier–Stokes problem which would result in a proof) and versions "C" and "D" (which would result in a disproof) to separate groups of agents.



“But AI writing code is a black box”

- But… PhD students/postdocs/any human writing code is a black box
- Academia has been built on black box code for decades
- Do we really think humans are better at parsing 1000s lines of text?
- Need to understand what code *does*, and know when we can *trust* code

vs



AI for developers

Developers – general + GitHub repos

- **Working with GitHub never changes** learn it and love it for *everything*
- **/init:** create "CLAUDE.md" file: repo overview/instructions for AI agent
- Give as much info as possible to start from
  - Papers, src text
- Give specific targets
  - Small, incremental review
  - Specific benchmarks and endpoints
- AI writes pretty good tests and can discover bugs
- Human should focus on code design + ideas
  - AI won't tell you a good idea on its own, and has bad taste
- Create a new branch for each experiment:
  - main = sacred
  - PR for new feature: human (and AI) code review, CI/CD tests pass

Developers – Setup

1. Settings for Claude
    a. Permissions
    b. Instructions

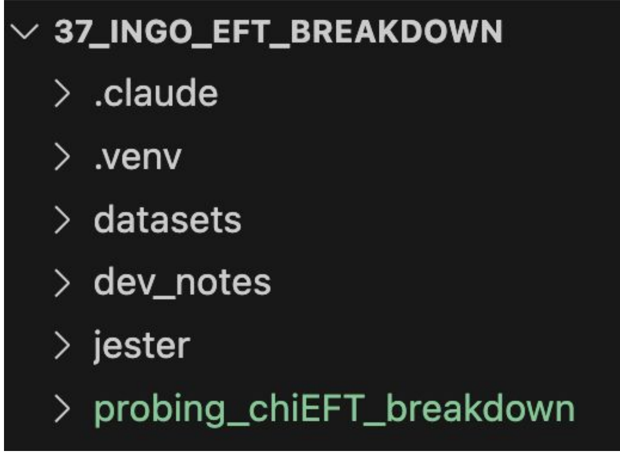
2. Python venv for the project

3. Subdirectories for extensive debugging/papers/…
    a. “Lab notes” and persistent memory

4. Code packages for which src code is crucial (jester, nmma, bilby,...)
    a. Point LLM model to specific scripts to guarantee code quality

5. Project-specific GitHub

1.
2.
3.
4.
5.



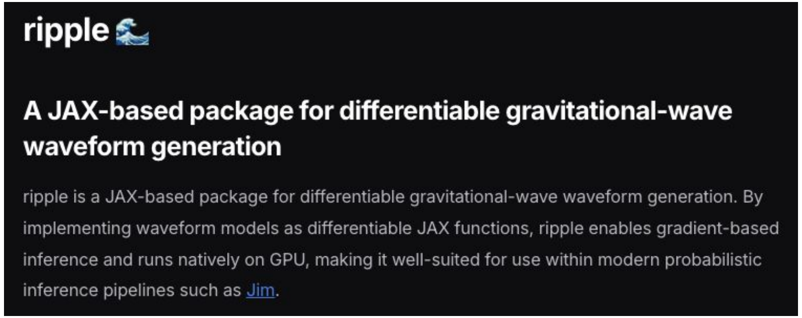
Developer case studies

- A failed example
- Spotting a bug #1
- Spotting a bug #2
- Conversing with source code

Developer – dos and don’ts

- Do
  - Small code snippets, well-defined scope, clear end goal
  - “There seems to be a bug with X when I run Y, Z,... investigate and find the root cause”

- Do not
  - Give a whole research project outline, expect a bug-free and correct outcome
  - “Translate this 1000+ line GW waveform from lalsuite in C to JAX. Be smart and do not make any mistake”



ripple 🌊

A JAX-based package for differentiable gravitational-wave waveform generation

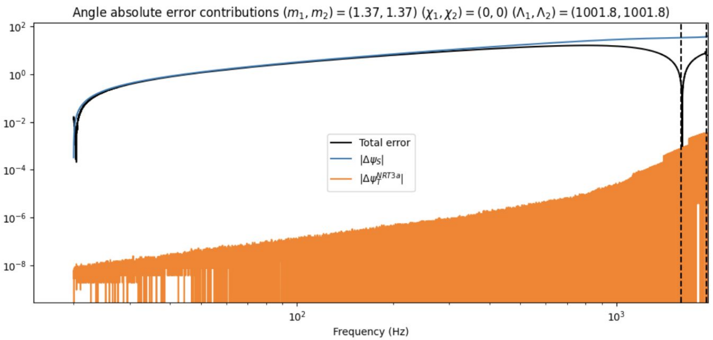
ripple is a JAX-based package for differentiable gravitational-wave waveform generation. By implementing waveform models as differentiable JAX functions, ripple enables gradient-based inference and runs natively on GPU, making it well-suited for use within modern probabilistic inference pipelines such as Jim.

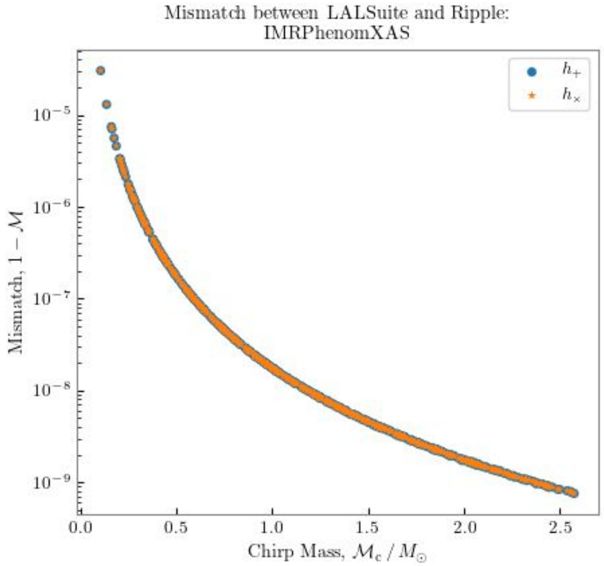
Developer case study – weighting the Sun

- GW waveforms translated into JAX
- Latest: IMRPhenomXAS_NRTidalv3 added by Robin Chan
- Long-standing bug… ~ October – December

NRTidalv3: zero-spin

Maybe I missed a contribution?
BBH baseline not correct?
Ripple vs LAL sign convention?





Developer case study – weighting the Sun

- Solved by Claude in **5 mins** by asking: *‘pls fix’*

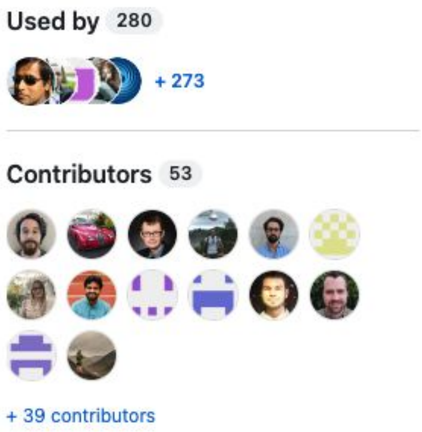
```
● Found it. The issue is a 1.58e-8 relative difference in the solar mass constant:

  - ripple MSUN: 1.988409902147e+30 kg
  - LAL MSUN:    1.988409870698e+30 kg

  This propagates to gt (G×M☉/c³), the fundamental time unit used throughout the waveform.
```

```diff

-MSUN = 1.988409902147041637325262574352366540e30  # kg
+MSUN = 1.9884098706980507e30  # kg (LAL_MSUN_SI)
 """Solar mass"""

-MRSUN = 1.476625038050124729627979840144936351e3
+MRSUN = 1476.6250380501247  # m (LAL_MRSUN_SI)
 """Geometrized nominal solar mass, m"""

-G = 6.67430e-11  # m^3 / kg / s^2
+G = 6.6743e-11  # m^3 / kg / s^2 (LAL_G_SI)
 """Newton's gravitational constant"""

 C = 299792458.0  # m / s
```

Developer case study – buggy bilby

- bilby = ***the*** #1 package for Bayesian inference on GW
- Used by the LVK collaboration and many research groups
- >50 human devs: so surely, there won't be any bugs!

- Let's see…



Developer case study – buggy bilby

- Can you spot a bug in the bilby relativ binning code?

Developer case study – buggy bilby

Thinks a few mins

> Can you spot a bug in the bilby relativ binning code?

Searched for 4 patterns, read 1 file, ran 3 shell commands

● Let me verify this with an actual reproduction rather than just reasoning about it.

Listed 1 directory, ran 2 shell commands

● Good, it's editable-installed against the local checkout. Let me write a minimal reproduction script.

Made 1 scratchpad edit +46, read 1 file, ran 1 shell command

● Let me fix the script based on this example.

Made 1 scratchpad edit +64, ran 1 shell command
⎿ Found 2 new diagnostic issues in 1 file (ctrl+o to expand)

● Confirmed — reproduced the exact crash I predicted, at exactly the line I flagged.

Yes — found and reproduced a real bug in RelativeBinningGravitationalWaveTransient.

Developer case study – buggy bilby

The mismatch: bin edges are chosen on the union of detector bands, but looked up on each detector's own (narrower) band

In `setup_bins`,

```
maximum_frequency = frequency_array[0]
minimum_frequency = frequency_array[-1]
for interferometer in self.interferometers:
    maximum_frequency = max(maximum_frequency, interferometer.maximum_frequency)
    minimum_frequency = min(minimum_frequency, interferometer.minimum_frequency)
```

takes the *loosest* (union) frequency band across all detectors: the smallest `minimum_frequency` and the largest `maximum_frequency` of any interferometer in the list. Bin edges (`self.bin_freqs`) are then placed anywhere in this wider, unioned band. But in `compute_summary_data`, each detector is processed against *its own*, potentially narrower, frequency mask,

```
mask = interferometer.frequency_mask
masked_frequency_array = interferometer.frequency_array[mask]
masked_bin_inds = []
for edge in self.bin_freqs:
    index = np.where(masked_frequency_array == edge)[0][0]
    masked_bin_inds.append(index)
```

Accelerated parameter estimation in Bilby with relative binning

Kruthi Krishna, Aditya Vijaykumar, Apratim Ganguly, Colm Talbot, Sylvia Biscoveanu Show All(8)

Dec 10, 2023

19 pages

e-Print: 2312.06009 [gr-qc]

View in: ADS Abstract Service

pdf cite claim reference search 56 citations

Developer case study – buggy bilby

The mismatch: bin edges are chosen on the union of detector bands, but looked up on each detector's own (narrower) band

In `setup_bins`,

```
maximum_frequency = frequency_array[0]
minimum_frequency = frequency_array[-1]
for interferometer in self.interferometers:
    maximum_frequency = max(maximum_frequency, interferometer.maximum_frequency)
    minimum_frequency = min(minimum_frequency, interferometer.minimum_frequency)
```

takes the *loosest* (union) frequency band across all detectors: the smallest `minimum_frequency` and the largest `maximum_frequency` of any interferometer in the list. Bin edges (`self.bin_freqs`) are then placed anywhere in this wider, unioned band. But in `compute_summary_data`, each detector is processed against *its own*, potentially narrower, frequency mask,

```
mask = interferometer.frequency_mask
masked_frequency_array = interferometer.frequency_array[mask]
masked_bin_inds = []
for edge in self.bin_freqs:
    index = np.where(masked_frequency_array == edge)[0][0]
    masked_bin_inds.append(index)
```

```
@article{Krishna:2023bug,
    author = "Krishna, Kruthi and Vijaykumar, Aditya and
    title = "{Accelerated parameter estimation in Bilby
    eprint = "2312.06009",
    archivePrefix = "arXiv",
    primaryClass = "gr-qc",
    month = "12",
    year = "2023"
}
```

Accelerated parameter estimation in Bilby with relative binning

Kruthi Krishna, Aditya Vijaykumar, Apratim Ganguly, Colm Talbot, Sylvia Biscoveanu Show All(8)

Dec 10, 2023

19 pages

e-Print: 2312.06009 [gr-qc]

View in: ADS Abstract Service

#nmma_questions on demand: N**LLM**MA

- Which kilonova models are available in NMMA?

```
From nmma/em/model.py, the model_parameters_dict (source of truth for what models exist and their parameters) – kilonova-specific ones:

SVD surrogate kilonova models (trained on radiative-transfer simulation grids):
- Bu2019lm – dynamical + wind ejecta, phi & theta (log10_mej_dyn, log10_mej_wind, KNphi, KNtheta)
- Bu2019lm_sparse – sparse variant, ejecta masses only
- Bu2019nsbh – NSBH-specific (log10_mej_dyn, log10_mej_wind, KNtheta)
- Bu2022mv – dynamical + wind with explicit ejecta velocities
- Bu2022Ye – adds dynamical electron fraction Yedyn
- Bu2023Ye – adds both Yedyn and Yewind
- Ka2017 – single-component (log10_mej, log10_vej, log10_Xlan)
- LANL2022, LANLTP1, LANLTP2, LANLTS1, LANLTS2 – LANL two-component ejecta models (dyn + wind, theta)

Analytic kilonova models:
- Me2017 – Metzger-style single-component (log10_mej, log10_vej, beta, log10_kappa_r)
- HoNa2020 – velocity-dependent opacity model with low/high-vej kappa split
- Arnett / Arnett_modified – bolometric radioactive-decay light curve (not kilonova-specific per se, but used for it)

Each entry's citation_dict (lines 134–172) points to the originating papers, e.g. Bu2019lm→arXiv:2002.11355, Ka2017→arXiv:1710.05463, Bu2022Ye/Bu2023Ye/Bu202

Note the dict also holds non-kilonova models (GRB TrPi2018, shock-cooling Piro2021, supernova AnBa2022_*/salt2, host-galaxy Sr2023) – I excluded those above

Cooked for 15s · done 6:53 PM
```

#nmma_questions on demand: N**LLM**MA

- What about fiesta?

```
Yes — there's a separate fiesta-based model path, distinct from the SVD/model_parameters_dict models I listed before. Fro

FiestaKilonovaModel (nmma/em/model.py:734) wraps fiesta.inference.lightcurve_model.BullaFlux (or BullaLightcurveModel if
"fiesta_grb" → FiestaGRBModel (line 846, wraps AfterglowFlux).

- Default model name: "Bu2026_MLP" (line 743/759 default arg) — the newest Bulla-grid MLP surrogate.
- The test suite (nmma/tests/fiesta_smoke.py:32-36) explicitly tries "Bu2026_MLP" first and falls back to "Bu2025_MLP" if
  fiesta-surrogates HuggingFace repo, just not hardcoded elsewhere in nmma (unlike the GitLab SVD models, fiesta model na
  <surrogate_dir>/KN/<model>/model).

So: Bu2025/Bu2026 aren't in the classic SVD list — they're fiesta surrogate names, only resolvable if the fiesta package
nuclear-multimessenger-astronomy/fiesta-surrogates repo containing KN/Bu2025_MLP/model or KN/Bu2026_MLP/model. There's no
surrogate repo.
```

#nmma_questions on demand: N**LLM**MA

In this venv: /Users/Woute029/Documents/Code/projects/NMMA_2026/presentation_AI/.venv/bin/python

Do I have fiesta installed?

Ran 1 shell command

No – fiesta is not installed in that venv (ModuleNotFoundError: No module named 'fiesta'). So Fie
environment. If you want to use them, you'd need to install it – it's not in the pyproject.toml e

Install it

Searched for 1 pattern, read 1 file

- fiestaEM is the PyPI package name (declared in the production extra in pyproject.toml:97). I'll install just that package into your venv.

Listed 2 directories, ran 3 shell commands

- This venv is managed by uv (no pip binary present). I'll install with uv pip install targeting this venv.

Ran 1 shell command

- Installed fiestaEM==0.2.1 (a JAX-based package – pulled in jax, flax, optax, etc.). Let's verify it imports correctly.

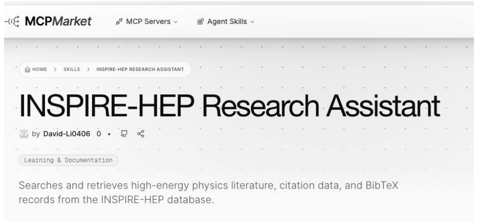
Ran 1 shell command

- fiesta now imports fine. Note: to actually use FiestaKilonovaModel/Bu2025_MLP/Bu2026_MLP, you still need the surrogate weights themselves – set NMMA_FIESTA_SURROGATES to a local checkout of the nuclear-multimessenger-astronomy/fiesta-surrogates HuggingFace repo (or pass surrogate_dir=... explicitly), since installing fiestaEM only gives you the inference code, not the trained models.

AI for research

AI for research?

- Meh… use with a lot of care, and only for shower thought brainstorms



- “My AI hallucinates references”: skill issue, truly
  - inspireHEP skill to query inspireHEP API
  - Bash “arxiv_src” command to download TeX src code

- Example: brainstorming on new samplers to use in NMMA

Beyond Nested Sampling: A Survey of Samplers for Multimessenger Bayesian Inference

Research notes for the NMMA / retreat discussion

Prepared with Claude Code (research assistant), for Thibeau Wouters

September 2026

→ Not good (no jackpot)

Prompt & results

- Samplers in multimessenger inference codes?
- Search other papers: what do they use?
- What about samplers in other communities?
- Use inspireHEP API to find papers
- Resulting PDF file here

```
I work on neutron stars, gravitational waves, multimessenger astrophysics, and a lot of Bayesian inference. We work a lot with stochastic samplers such as bilby, dynesty, pymultinest,... however, we believe those samplers are not
really optimal. You can find the src code here

/Users/Woute029/Documents/Code/projects/NMMA_2026/presentation_AI/nmma

We deal with high-dimensional parametrizations, and we also have slow likelihoods that are mainly dominated by the GW waveform, and perhaps the EOS if we solvee the TOV equations on the fly. We ahve different sources: GW, EM (GRB and
kilonovae), and EOS,...

Can you search
- which other papers are out there that are like NMMA, in the sense that they perform sampling on multimessenger sources?
- is their code open source?
- which samplers do they use?

However, the multimessenger community might be slow with catching up. Can we learn from other communities that also had to do Bayesian inference in high-dim spaces, whcih tools do they use, and do you see opportunities for
cross-culture works and can we adopt their research methods?

Write down your findings and thoughts and recommendations in a LaTeX doc, and compile PDF, here: /Users/Woute029/Documents/Code/projects/NMMA_2026/presentation_AI/retreat/thibeau_presentation_AI_demo/research_demo/

Importantly, use inspireHEP API to find sources and document them in references.bib to back up your claim. For interesting papers, use arxiv_src to fetch the src code to verify claims (eg on performance and how it compares to NMMA if
possible but also on the samplers and numerical methods they use).
```

AI for writers

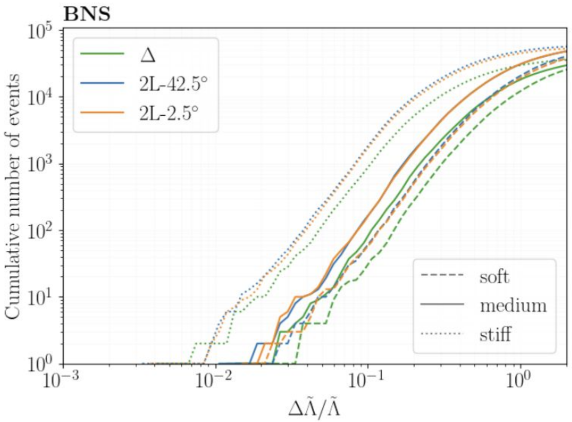
NO!!!

- Except for writing that is actually coding
    - Proofreading and fixing typos (= language ‘bugs’)
    - Buggy TeX code
    - Plotting code… an example

Tim Dietrich 9:43 AM

Hi Thibeau, can I bother you again and ask for quick help; I am using francesco's notebook, you have been able to plot this?

Screenshot 2026-09-02 at 09.43.49.png



In [21]:
```python
import numpy as np
import matplotlib.pyplot as plt
import sys, os
import h5py
import corner
from matplotlib.lines import Line2D

# rcParams
plt.rcParams["axes.axisbelow"] = True  # Set the grid beneath the curves
plt.rcParams["text.usetex"] = True     # Use the tex fonts
plt.rcParams["font.family"] = "serif"  # Use the "serif" tex font
plt.rcParams["font.size"] = 14         # Set the default font size to 16
plt.rcParams["text.latex.preamble"] = r"\usepackage{bm}"
import os
os.environ["PATH"] += os.pathsep + "/Library/TeX/texbin"
panel_size=(6, 4)
wspace_in=0.6
left_in=0.9
right_in=0.25
bottom_in=0.75
top_in=0.45
```

Horizons

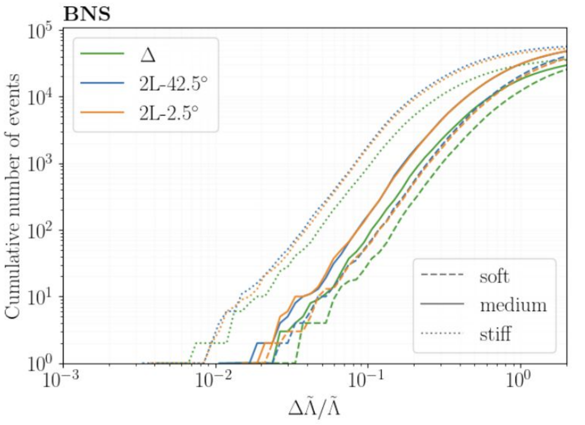
In [2]:
```python
HORDIR = '../results/gwfast_results_horizon/'
horread_T, horread_Lmis, horread_Lpar = {}, {}, {}
with h5py.File(HORDIR + 'inference_horizon_T_q1chi0_XHM.h5', 'r') as f:
    for key in f.keys():
        horread_T[key] = f[key][()]
with h5py.File(HORDIR + 'inference_horizon_Lmis_q1chi0_XHM.h5', 'r') as f:
    for key in f.keys():
        horread_Lmis[key] = f[key][()]
with h5py.File(HORDIR + 'inference_horizon_Lpar_q1chi0_XHM.h5', 'r') as f:
    for key in f.keys():
        horread_Lpar[key] = f[key][()]
horizons_T = np.loadtxt('/Users/francescoiacovelli/Library/CloudStorage/OneDrive-
horizons_Lmis = np.loadtxt('/Users/francescoiacovelli/Library/CloudStorage/OneDri
horizons_Lpar = np.loadtxt('/Users/francescoiacovelli/Library/CloudStorage/OneDri
```

~1000 lines

Tim Dietrich 9:43 AM

Hi ~~Thibeau~~, can I bother you again and ask for quick help; I am using francesco's notebook, you have been able to plot this?

Screenshot 2026-09-02 at 09.43.49.png



In [21]:
```python
import numpy as np
import matplotlib.pyplot as plt
import sys, os
import h5py
import corner
from matplotlib.lines import Line2D

# rcParams
plt.rcParams["axes.axisbelow"] = True  # Set the grid beneath the curves
plt.rcParams["text.usetex"] = True     # Use the tex fonts
plt.rcParams["font.family"] = "serif"  # Use the "serif" tex font
plt.rcParams["font.size"] = 14         # Set the default font size to 16
plt.rcParams["text.latex.preamble"] = r"\usepackage{bm}"
import os
os.environ["PATH"] += os.pathsep + "/Library/TeX/texbin"
panel_size=(6, 4)
wspace_in=0.6
left_in=0.9
right_in=0.25
bottom_in=0.75
top_in=0.45
```

Horizons

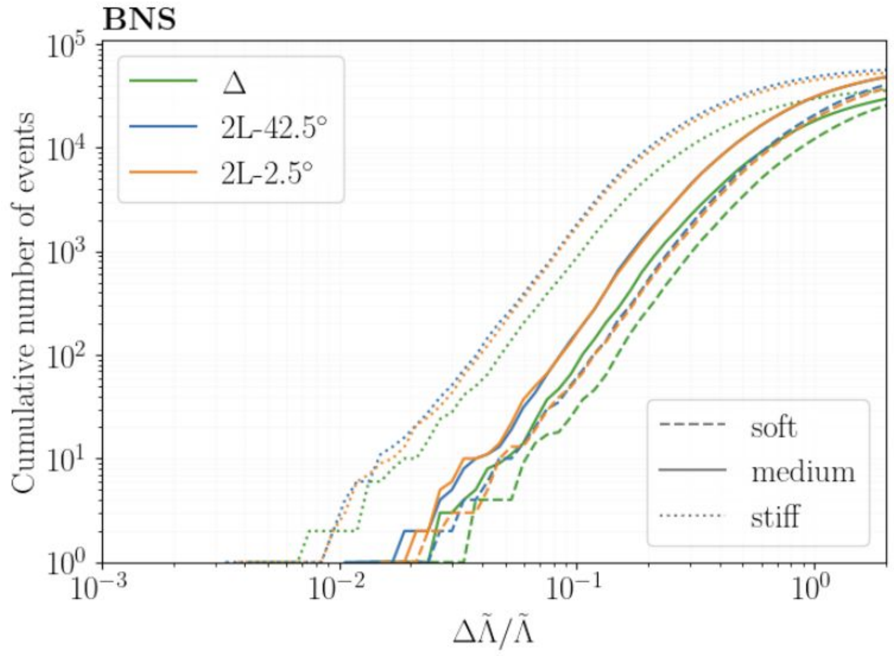
In [2]:
```python
HORDIR = '../results/gwfast_results_horizon/'
horread_T, horread_Lmis, horread_Lpar = {}, {}, {}
with h5py.File(HORDIR + 'inference_horizon_T_q1chi0_XHM.h5', 'r') as f:
    for key in f.keys():
        horread_T[key] = f[key][()]
with h5py.File(HORDIR + 'inference_horizon_Lmis_q1chi0_XHM.h5', 'r') as f:
    for key in f.keys():
        horread_Lmis[key] = f[key][()]
with h5py.File(HORDIR + 'inference_horizon_Lpar_q1chi0_XHM.h5', 'r') as f:
    for key in f.keys():
        horread_Lpar[key] = f[key][()]
horizons_T = np.loadtxt('/Users/francescoiacovelli/Library/CloudStorage/OneDrive-
horizons_Lmis = np.loadtxt('/Users/francescoiacovelli/Library/CloudStorage/OneDri
horizons_Lpar = np.loadtxt('/Users/francescoiacovelli/Library/CloudStorage/OneDri
```

~1000 lines

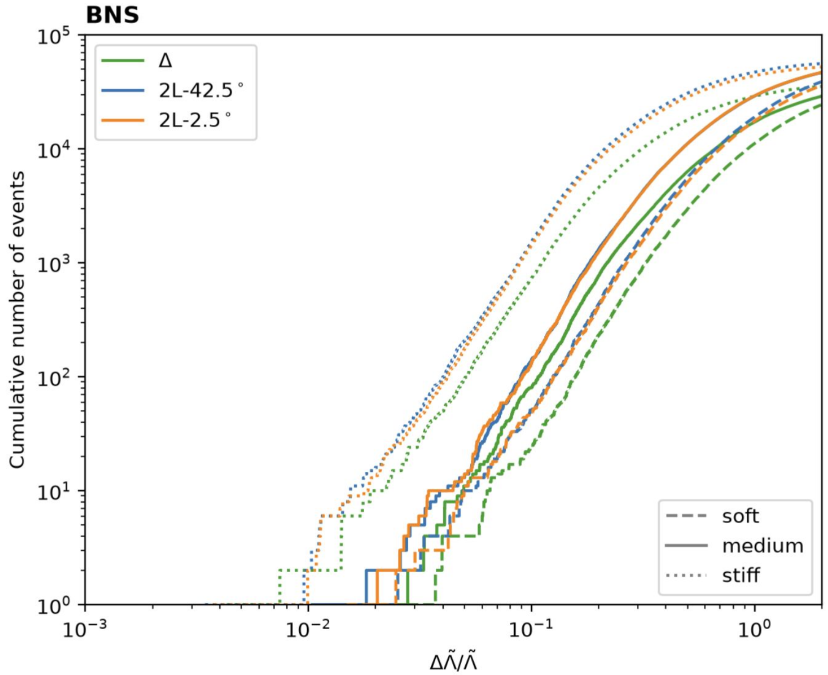
```
Claude Code v2.1.258
Sonnet 5 with high effort · Claude Pro
~/Documents/Code/projects/[redacted]

In ./tim , make a new script called make_fisher_plots.py that recreates this figure

[Image #1]

Directly from the ./datasets that we have: using the /Users/Woute029/Documents/Code/projects/[redacted]_results and the injection EOSs.

Make the following dictionary for translating labels: APR4 is "soft", H4 is "stiff" and SFHo is "medium"

Make the BNS catalog for Delta Lambda Tilde as you see in the figure
  [Image #1]
```

```
● This matches the reference figure closely – same curve shapes, ordering, and knee structure. Created tim/make_fisher_plots.py, which loads each
  gwfast_results_<config>_BNS_<EOS>_snrth-8.h5 + its matching injection catalog from ./datasets, computes err_LambdaTilde / LambdaTilde_true per detected
  source, and plots cumulative event counts on log-log axes (green/blue/orange = Δ/2L-42.5°/2L-2.5°, dashed/solid/dotted = soft/medium/stiff for
  APR4/SFHo/H4). Output saved to tim/figures/bns_delta_lambda_tilde_cumulative.png.

* Worked for 1m 33s   done 9:56 AM
```

Input image



Output image

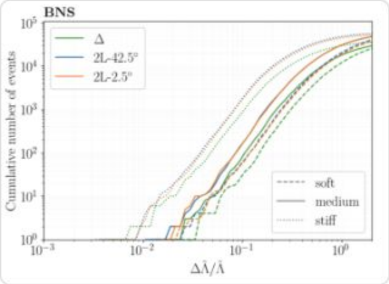


Tim Dietrich 9:43 AM

Hi Thibeau, can I bother you again and ask for quick help; I am pushing a bit since we were ask to provide a finalized draft of the sections by the end of the week, so that I focus today on writing things up so that I can ask for feedback

using francesco's notebook, you have been able to plot this?

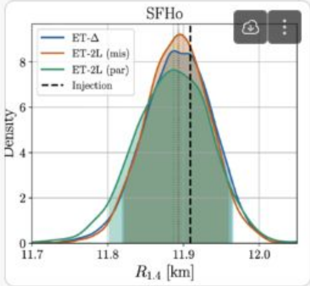
Screenshot 2026-09-02 at 09.43.49.png



If so, can you make a plot just for "medium" which is SFHo and show \Delta R/R using Delta R/R = 1/6 Delta Lambda_tilde/Lambda_tilde

9:45 and I would need the jester result plot with the label "medium" on top

Screenshot 2026-09-02 at 09.45.22.png



because we were asked to use medium/soft/stiff

ahh and in the top right, can you add the radius confidence intervals in the same colors as the detector types

Thibeau Wouters 10:19 AM

Price: one beer per PDF 🙂

6 files ▾ | Download all

- bns_delta_lambda_tilde_cumulativ... PDF
- bns_delta_lambda_tilde_cumulativ... PDF
- bns_delta_radius_cumulative_all.pdf PDF
- bns_delta_radius_cumulative_medi... PDF
- master_radius_1p4_snr100_all.pdf PDF
- master_radius_1p4_snr100_mediu... PDF

🍺 🍺 🍺 🍺 🍺 🍺

Closing thoughts

Summary

- AI accelerates code development
- Humans
  - Bigger picture
  - Ideas and insights
  - Core understanding of the problem
- AI agents
  - Crunch small tasks very fast
  - Dig into parallel rabbit holes
- Invest in critical thinking, understanding *how* research should be done
- Know how/when to use AI agent, with care

"Using AI is shattering my belief on how research is done"

- The beast is unleashed now and there is no way to tame it

“Using AI is shattering my belief on how research is done”

- The beast is unleashed now and there is no way to tame it,... so wanna try?
    - 3 passes for 1 week to share

AI for NMMA retreat

AI for NMMA retreat

- Scouting the Git issues and PRs, compiled into a PDF document [here]
- We (humans) should focus on brainstorms about the bigger picture!!!

NMMA Codebase Review 1 / 9

NMMA Codebase Review — Retreat Prep

Generated by Claude Code

2026-08-23 *Research only — no code changed, nothing pushed/merged*

LET's GO!!!